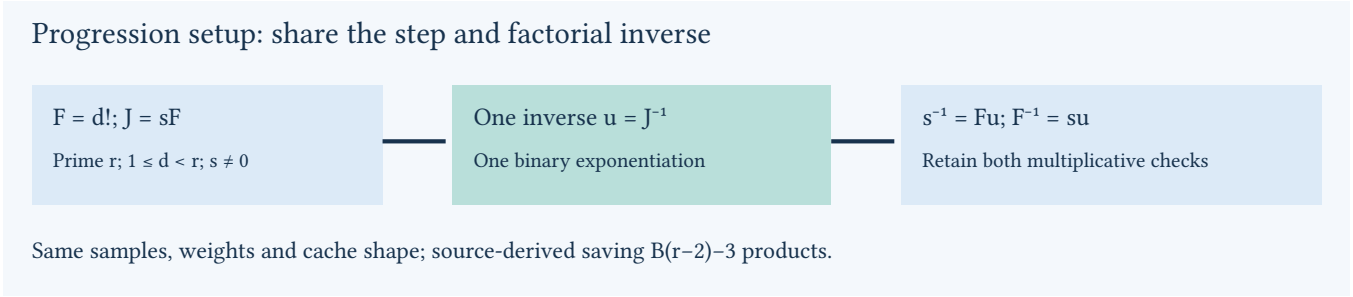
Shared progression setup inverse

T-1889 · Prime-field coefficient construction · 2026-11-08

Progression setup now obtains the grid-step and factorial inverses from one product inverse. Both existing verification multiplications remain. This applies at each setup reached by the interval engine, including the coefficient-stream path.



Identity. For prime r, nonzero step s and 1 ≤ d < r, F=d! is nonzero. Set u=(sF)^(r−2). Then Fu=s^{-1} and su=F^{-1}. Substitution preserves the samples, inverse-factorial recurrence and Lagrange weights. No new vector or retained cache field is introduced.

Source-derived arithmetic change. The binary exponentiation helper uses B(n)=bit_length(n)+popcount(n) multiplication calls, including its final square. Replacing two inverses by one adds three multiplications, yielding B(r−2)−3 fewer scalar multiplication calls per setup. This is 2 at r=7, 10 at r=97, 13 at r=257 and 29 at r=65537. These are derived code counts, not runtime observations.

Qualification. The new controls cover 234 setups, 2,556 sample/weight identities and 151,020 evaluation values, with 45 setups at degree r−1. They require one Fermat-inverse entry per setup and retain five invalid-domain controls. Zero polynomials and all first points in the selected small fields are included. Native receipt records the inherited regression result separately.

Evidence. Reversible source edits and compiler dependency hashes bind the implementation. The inherited recovery checker is compared byte for byte with preserved output. Producer assertions remain unsigned pending independent signed verification.

Scope and next step. Prime-field preconditions remain caller obligations. No comparative width panel or cloud/timing workload ran. Future width comparisons must bind this same progression implementation in every arm. The larger source/coefficient cost accounting remains separate from this local arithmetic reduction.